

Implementación de Red de Interdependencia

Marcos Laserna Moratalla

27-12-2025

Contents

1	Funciones de Correlación Wavelet	1
2	Estudio de Bonos Verdes (2021-2024)	2
2.1	Procesamiento de datos	2
2.2	Visualización de la Red	3
3	Estudio de Acciones Varias (2021-2024)	4
3.1	Procesamiento de datos	4
3.2	Visualización de la Red	5
4	Dendrograma	7

1 Funciones de Correlación Wavelet

```
fwav <- function(sig1, sig2, scales, dt = 1) {  
  x <- as.numeric(sig1); y <- as.numeric(sig2)  
  
  # Transformada Wavelet  
  cwt_x_obj <- cwt_wst(x, dt = dt, scales = scales, wname = "MORLET", wparam = 6,  
    border_effects = "BE", makefigure = FALSE)  
  cwt_y_obj <- cwt_wst(y, dt = dt, scales = scales, wname = "MORLET", wparam = 6,  
    border_effects = "BE", makefigure = FALSE)  
  
  # Transposición  
  cwt_x <- t(cwt_x_obj$coefs)  
  cwt_y <- t(cwt_y_obj$coefs)  
  
  # Cálculo de coeficientes del paper  
  S_Wxy <- cwt_x * Conj(cwt_y)  
  S_PowX <- Mod(cwt_x)^2  
  S_PowY <- Mod(cwt_y)^2  
  
  # Suavizado S (preguntar duda sobre escala s, mi suavizado es lineal)  
  # porque son medias móviles  
  
  for(k in 1:length(scales)) {
```

```

win <- round(3 * scales[k]); if(win < 1) win <- 1
S_Wxy[k, ] <- caTools::runmean(S_Wxy[k, ], k = win, align = "center", endrule = "mean")
S_PowX[k, ] <- caTools::runmean(S_PowX[k, ], k = win, align = "center", endrule = "mean")
S_PowY[k, ] <- caTools::runmean(S_PowY[k, ], k = win, align = "center", endrule = "mean")
}

# Parámetros
R2 <- (Mod(S_Wxy)^2) / (S_PowX * S_PowY + 1e-10)
Phi <- Arg(S_Wxy)

return(list(R2 = R2, Phi = Phi))
}

```

2 Estudio de Bonos Verdes (2021-2024)

2.1 Procesamiento de datos

```

map <- data.frame(
  Paper_Name = c("GOLD", "GRNENEF", "GRNFUEL", "GRNGB", "GRNPOL", "GRNSOLAR",
    "GRNTRN", "GRNWIND", "HYBOND", "IGBOND", "MSCI", "OIL", "SPGBI", "TREAS"),
  Yahoo_Ticker = c("GC=F", "ICLN", "PLUG", "VNQ", "PBD", "TAN", "DRIV", "FAN",
    "HYG", "LQD", "URTH", "CL=F", "BGRN", "IEF")
)

prices_list <- lapply(setNames(map$Yahoo_Ticker, map$Paper_Name), function(ticker) {
  tryCatch({
    tmp <- getSymbols(ticker, src = "yahoo", from = "2020-01-01",
      to = "2024-01-01", auto.assign = FALSE)
    Ad(na.omit(tmp))
  }, error = function(e) NULL)
})

# Limpieza y ajustes de nombres
portfolio <- na.omit(do.call(merge, prices_list))
colnames(portfolio) <- names(prices_list)

returns <- diff(log(portfolio))[-1]
data_mat <- scale(returns)
fin <- colnames(data_mat)
n <- ncol(data_mat)

scales <- pow2scales(c(2, 128, 4))

f_factor <- 1.033

# Mayor a un mes
idx_band_c <- which(f_factor * scales > 30)

# Obtenemos matrices de correlación wavelet
adj_R2 <- matrix(0, n, n, dimnames = list(fin, fin))

```

```

adj_Phi <- matrix(0, n, n, dimnames = list(fin, fin))

for(i in 1:n) {
  for(j in 1:n) {
    if(i != j) {
      res <- fwav(data_mat[,i], data_mat[,j], scales)

      adj_R2[i,j] <- mean(res$R2[idx_band_c, ], na.rm = TRUE)
      adj_Phi[i,j] <- mean(res$Phi[idx_band_c, ], na.rm = TRUE)
    }
  }
}
diag(adj_R2) <- 1

```

2.2 Visualización de la Red

```

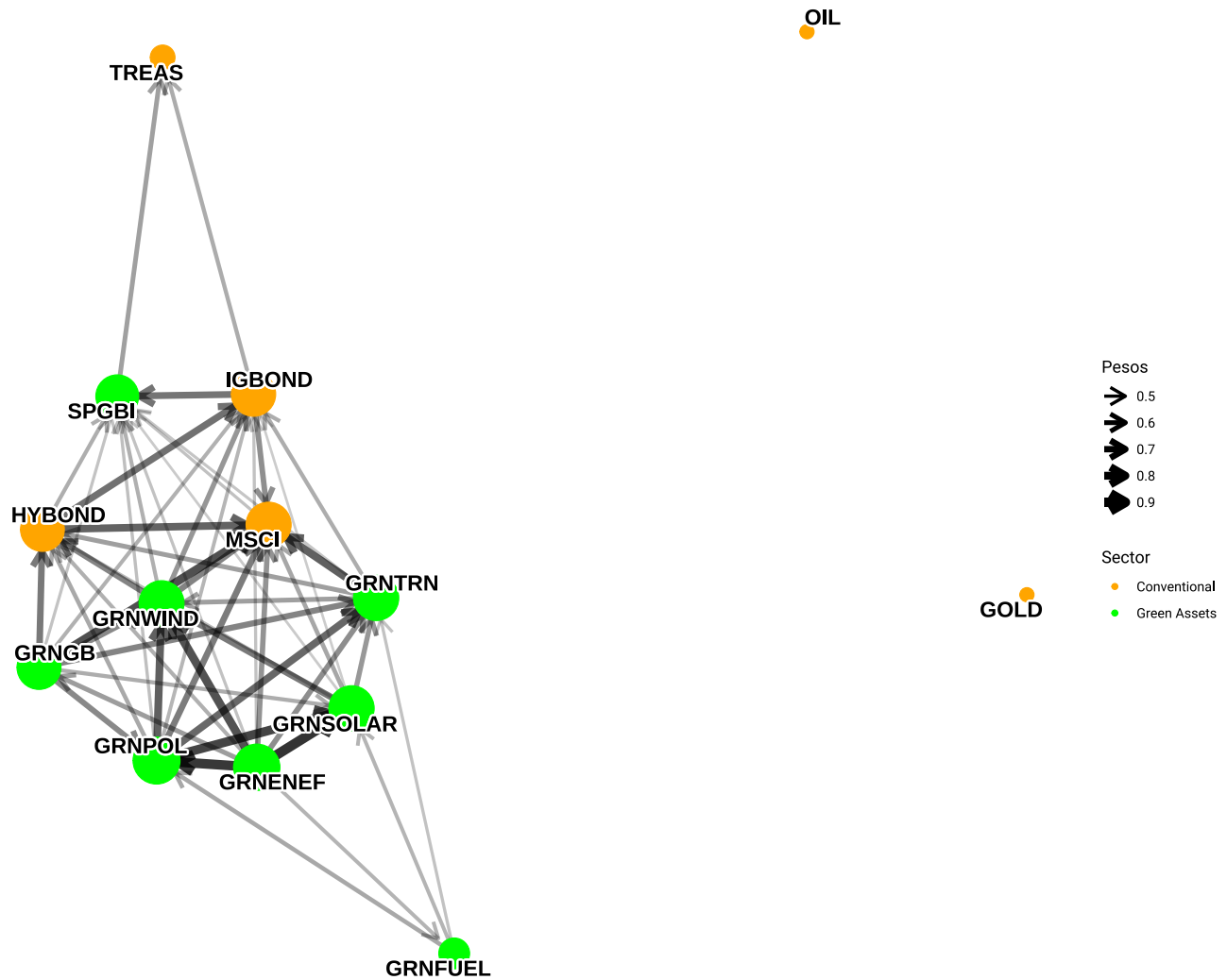
threshold <- 0.42
edges <- data.frame(from=character(), to=character(), weight=numeric(), phase=numeric())
for(i in 1:(n-1)) {
  for(j in (i+1):n) {
    w <- adj_R2[i,j]
    if(!is.na(w) && w > threshold) {
      edges <- rbind(edges, data.frame(from=fin[i],
                                         to=fin[j], weight=w, phase=adj_Phi[i,j]))
    }
  }
}

net <- graph_from_data_frame(edges, vertices=fin, directed=TRUE)
green_list <- c("SPGBI", "GRNENEF", "GRNFUEL", "GRNGB", "GRNPOL", "GRNSOLAR", "GRNTRN", "GRNWIND")
V(net)$Sector <- ifelse(V(net)$name %in% green_list, "Green Assets", "Conventional")
V(net)$Degree <- strength(net, mode="all"); V(net)$Degree[V(net)$Degree==0] <- 0.5

set.seed(555)
layout <- create_layout(net, layout = "fr", niter=2000)

ggraph(layout) +
  geom_edge_link(aes(width = weight, alpha = weight), color = "black",
                 arrow = arrow(length = unit(4, 'mm')), end_cap = circle(5, 'mm')) +
  scale_edge_width(range = c(0.8, 3), name="Pesos") +
  scale_edge_alpha(range = c(0.2, 0.8), guide="none") +
  geom_node_point(aes(color = Sector, size = Degree))+
  scale_size(range = c(4, 14), guide = "none")+
  scale_color_manual(values = c("Conventional" = "orange", "Green Assets" = "green")) +
  geom_node_text(aes(label = name), repel=TRUE, fontface="bold", size=5, bg.color="white", bg.r=0.15) +
  expand_limits(x = c(-1.5, 1.5), y = c(-1.5, 1.5)) +
  theme_graph() +
  theme(legend.position = "right", text = element_text(family = "roboto"))

```



3 Estudio de Acciones Varias (2021-2024)

3.1 Procesamiento de datos

```
map_2 <- data.frame(
  Sector = c(rep("Tecnología", 5), rep("Crypto", 4), rep("Energía", 4),
    rep("Farmacéutica y Defensa", 5), rep("Índices", 5), rep("Divisas", 6)),
  Yahoo_Ticker = c("NVDA", "MSFT", "AAPL", "GOOGL", "TSLA", "BTC-USD", "ETH-USD",
    "SOL-USD", "BNB-USD", "XOM", "CVX", "BA", "CAT", "LLY", "JNJ",
    "PFE", "MRK", "ABBV", "SPY", "QQQ", "^VIX", "GLD", "TLT", "UUP",
    "EUR=X", "GBP=X", "JPY=X", "CHF=X", "AUD=X"),
  Label = c("Nvidia", "Microsoft", "Apple", "Google", "Tesla", "Bitcoin", "Ethereum",
    "Solana", "Binance", "Exxon", "Chevron", "Boeing", "Caterpillar", "EliLilly",
    "J&J", "Pfizer", "Merck", "AbbVie", "S&P500", "Nasdaq", "VIX", "ORO",
    "Bonos20A", "Dolar(IDX)", "Euro", "Libra", "Yen", "FrancoSuizo", "AudDolar")
)
```

```

prices_list <- list()
for(i in 1:nrow(map_2)){
  tryCatch({
    tmp <- getSymbols(map_2$Yahoo_Ticker[i], src="yahoo",
                      from="2021-01-01", to="2024-01-01", auto.assign=FALSE)
    prices_list[[map_2$Label[i]]] <- Ad(na.omit(tmp))
  }, error=function(e) NULL)
}

# Para que los nombres se vean bien
portfolio_2 <- na.omit(do.call(merge, prices_list))
colnames(portfolio_2) <- names(prices_list)

data_2 <- scale(diff(log(portfolio_2))[-1])
fin2 <- colnames(data_2)
n_2 <- ncol(data_2)

scales <- pow2scales(c(2, 128, 4))
f_factor <- 1.033

# Mínimo de un mes
idx_band_c <- which(f_factor * scales > 30)

# Calculo de matrices
adj_R2_2 <- matrix(0, n_2, n_2, dimnames=list(fin2, fin2))
adj_Phi_2 <- matrix(0, n_2, n_2, dimnames=list(fin2, fin2))

for(i in 2:n_2) {
  for(j in 1:(i-1)) {
    res <- fwav(data_2[,i], data_2[,j], scales)
    adj_R2_2[i,j] <- adj_R2_2[j,i] <- mean(res$R2[idx_band_c, ], na.rm=TRUE)
    adj_Phi_2[i,j] <- mean(res$Phi[idx_band_c, ], na.rm=TRUE)
    adj_Phi_2[j,i] <- -adj_Phi_2[i,j]
  }
}
diag(adj_R2_2) <- 1

```

3.2 Visualización de la Red

```

threshold_2 <- 0.42

edges_2 <- data.frame(from=character(), to=character(), weight=numeric())

for(i in 1:(n_2-1)) {
  for(j in (i+1):n_2) {
    w <- adj_R2_2[i,j]
    if(!is.na(w) && w > threshold_2) {
      edges_2 <- rbind(edges_2, data.frame(from=fin2[i], to=fin2[j], weight=w))
    }
  }
}

net_2 <- graph_from_data_frame(edges_2, vertices=fin2, directed=TRUE)
indices <- match(V(net_2)$name, map_2$Label)

```

```

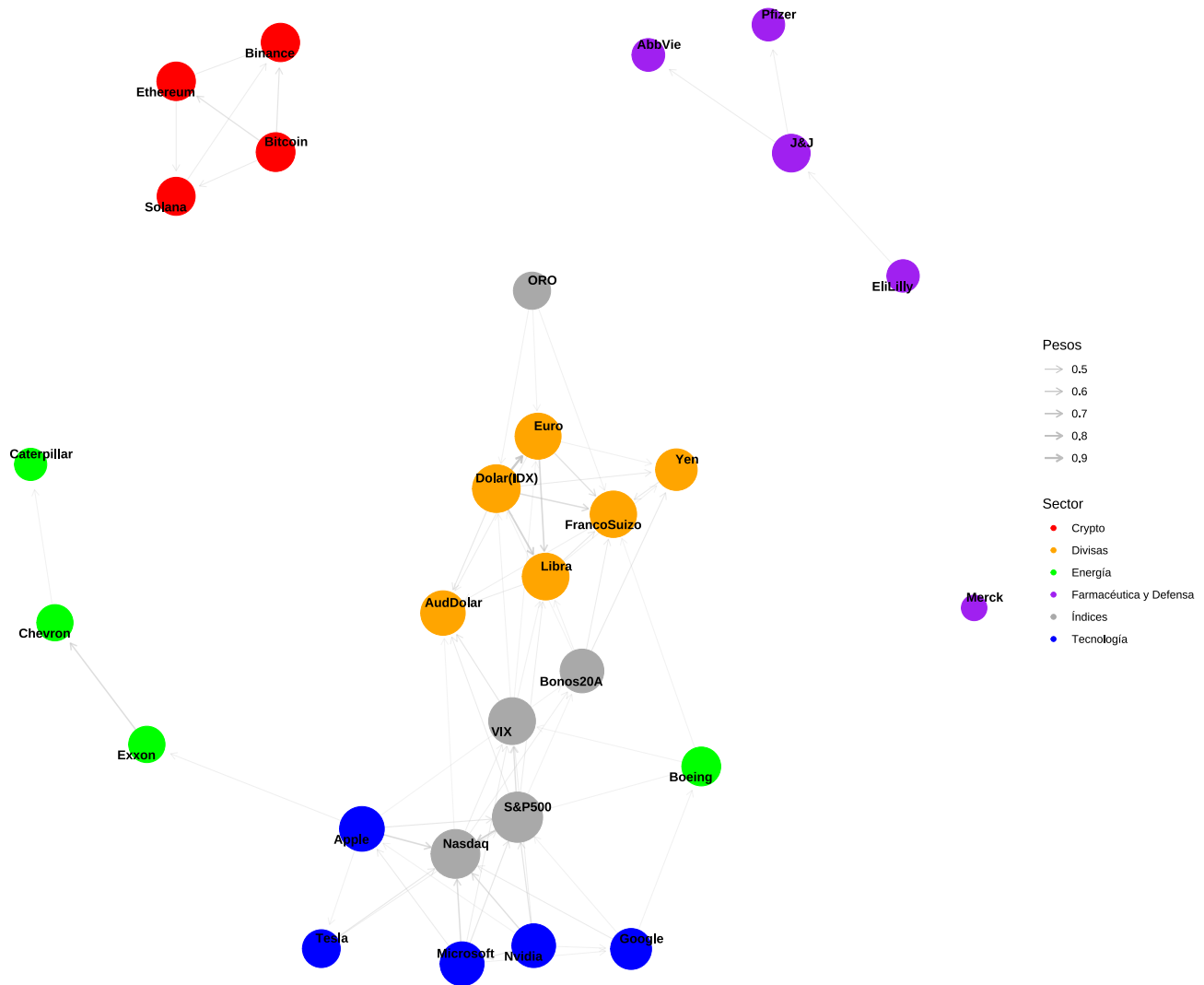
V(net_2)$Sector <- map_2$Sector[indices]
V(net_2)$Degree <- strength(net_2, mode="all")

set.seed(555)
ggraph(net_2, layout = "fr") +
  geom_edge_link(aes(width = weight, alpha = weight), color = "grey",
    arrow = arrow(length = unit(2, 'mm')), end_cap = circle(7, 'mm')) +
  geom_node_point(aes(color = Sector, size = Degree)) +

  scale_edge_width(range = c(0.2, 0.8), name = "Pesos") +
  scale_edge_alpha(range = c(0.2, 0.8), guide = "none") +
  scale_size(range = c(9, 18), guide = "none") +

  geom_node_text(aes(label = name), repel=TRUE, fontface="bold", size=3.5) +
  theme_graph() +
  scale_color_manual(values = c("Tecnología"="blue", "Crypto"="red", "Energía"="green",
    "Farmacéutica y Defensa"="purple", "Índices"="darkgrey", "Divisas"="orange"))

```



4 Dendrograma

```
dist_matrix_2 <- as.dist(1 - adj_R2_2)
hc_macro <- hclust(dist_matrix_2, method = "ward.D2")

colores_clusters <- c("red", "blue", "green", "purple", "orange")

fviz_dend(hc_macro, k = 5, cex = 0.8,
          k_colors = colores_clusters,
          type = "circular", rect = TRUE,
          main = "Clusters Estratégicos Detectados") +
  theme(plot.title = element_text(hjust = 0.5, face="bold", family="roboto")) +
  theme(legend.position = "none")
```

-1.0

-0.5

0.0

0.5

1.0

